

Investigating Simple Privacy Indicators for Supporting Users when Installing New Mobile Apps

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ABSTRACT

Mobile devices have become daily companions for millions of users. They have access to privacy-sensitive data about their users which stresses the importance of privacy. Users have to make privacy-related decisions already before app installation because once installed, apps can access potential privacy-sensitive data. In this work-in-progress, we present an in-depth investigation of privacy indicator visualizations for mobile app stores. We report the results of two consecutive user studies in which we investigate 1) visual depiction, 2) score, and 3) monetary value of collected data. Our studies reveal that a visual depiction by a privacy meter were easiest to understand for users, scores were easiest to spot, and monetary value was most difficult to interpret and requires further investigation.

CCS CONCEPTS

• Human-centered computing → Empirical studies in HCI.

KEYWORDS

information privacy, privacy indicator, mobile privacy

ACM Reference Format:

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1 INTRODUCTION & RELATED WORK

Smartphones and other mobile devices have become daily companions for a plethora of users. The devices can offer functionality

based on integrated sensors, like cameras or microphones, and data analysis entered by their users. Consequently, mobile devices can have access to sensitive information, making the privacy of data crucial [4].

By definition, privacy is an individual concept meaning that users should be able to control the circumstances under which third parties process and store personal data [30]. Existing studies found that users that have privacy concerns do not necessarily consider privacy while installing new apps on their devices [9, 16, 25]. However, providing privacy-related information during decision making can nudge users to choose apps that match their individual privacy preferences [10, 13, 16–18].

There are several possibilities to present privacy-related information to users: privacy policies [2], privacy labels [15], privacy information platforms [20], or privacy indicators [5, 19, 23]. Although privacy policies are rich in terms of information, users often feel overwhelmed by the large amount of information and the time required for reading and understanding [1]. As a consequence, users often do not read privacy policies [4, 15, 21] or other privacy-related information [9]. Privacy labels are visualizations that summarize information from privacy policies to make it more accessible for users [8, 15, 17, 22]. Privacy information platforms are websites that accumulate privacy-related information about apps. These platforms are located outside the app stores. Hence, users have to search for them actively.

Finally, privacy indicators accumulate privacy-related information into one score, visualization, or indicator [19]. These indicators are promising because users do not require much effort to get an impression of an app's privacy properties. Studies have even shown that privacy indicators guide users to installing apps that match the user's privacy attitudes [12, 27]. However, placement, design, and timing of presenting the rating are crucial [7].

In this work-in-progress paper, we present an investigation of different privacy visualizations for privacy indicators. We first conducted an online survey with different indicator visualization options with 119 participants. In this study, we specifically investigated three categories of visualizations: 1) visual depiction, 2) privacy score, and 3) monetary value of collected data. Using the results from the online study, we chose one visualization possibility for each category. We implemented the three visualization possibilities as a badge for the Google Play Store.

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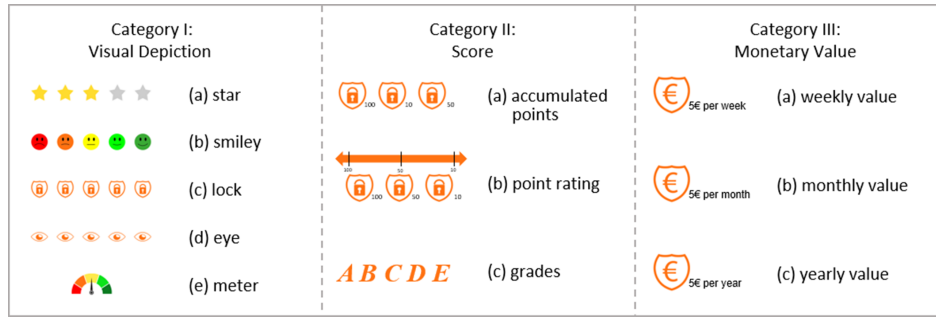


Figure 1: Overview indicators developed in Step I and investigated in Step II.

The implementation was used during a lab study with 20 participants in which we evaluated the three indicator categories and no indicator as a baseline. Overall, the participants noticed the indicators and used the information provided by them for choosing an app. The visual depiction by a privacy meter was easiest to understand since it did not require previous knowledge. Based on our study results, we provide specific recommendations for indicator design and motivate the next steps for future work.

2 METHODOLOGY OVERVIEW

In this section, we describe our underlying methodology for the investigation of privacy indicators for mobile apps. Overall, we followed three main steps:

- (1) *Step I (Indicator Development)*: In the first step, we developed three categories of privacy indicators based on related work [3, 23, 24]. Each category of privacy indicators contains several visualization possibilities.
- (2) *Step II (Survey Study)*: In the second step, we evaluated the privacy indicators from Step I in an online survey ($N = 119$). We aimed to investigate to what extent the privacy indicators were correctly understood by users and which ones they preferred. The preferred visualization possibilities of the privacy indicators from each category served as the basis for Step III.
- (3) *Step III (Lab Study)*: In the lab study, the participants were asked to complete app selection tasks. Then, we interviewed them to investigate how they perceive the various privacy indicators in an app store and to which extent the indicators influence their decision to install an app.

All studies reported in this paper were conducted in line with the recommendations from the ethics committee at our university. Before each study, participants were shown a consent form containing information about the goal, procedure, and the data protection policy. Participants could withdraw at any point during the study without providing any reason, and we informed them that in this case all collected data was deleted. Data collection and storage was in line with national and regional data protection laws. There was no compensation for taking part in the studies. The studies were conducted before the outbreak of COVID-19.

In the following sections, goal, procedure, and the results of the three steps are described in detail.

3 STEP I: INDICATOR DEVELOPMENT

In Step I of the study, we developed three categories of privacy indicators building on related work and the recommendation of Rajivan et al. [23]. The categories are (1) visual depiction, (2) score, and (3) monetary value. Each category contains different visualization possibilities (see Figure 1).

The category *visual depiction* consists of five visualization possibilities (star, smiley, lock, eye, and meter). Stars are a popular rating system used in websites and in app stores to indicate user ratings. Smileys are popular emoticons used while messaging to convey emotions. Eyes were used by Benton et al. [3] and Rajivan et al. [23] to communicate privacy risks. A lock symbol is commonly associated with physical security and computer security and gives an impression of privacy. It was used as a risk scale by Rajivan et al. [23]. In our study, a high number of eyes and a high number of locks were used to represent high levels of privacy. The meter, which has already been studied in other contexts, such as user authentication [6, 24, 28], has been adapted for the privacy context.

The category *score* in general indicates privacy by providing a score to the user. This score can be realized by the following three representations: (a) accumulated points, (b) point rating, and (c) grades. The scientific literature offers options to calculate scores. The score number or grade can be calculated using the availability of the privacy policy link, the number of permissions stated by the developer vs. the real permissions requested by the app, and the number of mobile trackers. Using the approach of Hatamian et al. [14] and Lumen Privacy Monitor [29], we can get the information to generate a number that indicates high or low privacy.

The category *monetary value* depicts the monetary value of the collected data. Related work demonstrated that users are willing to pay for premium apps based on privacy indicators [11, 27]. The monetary value of the collected data aims to offer users a value of their data. Here, we differentiate between three different visualizations. These specify usage time-frames of one week, one month, and one year.

4 STEP II: SURVEY STUDY

After developing the three categories of privacy indicators and visualization possibilities, we evaluated them in an online survey with 119 participants to investigate comprehensibility and preference.

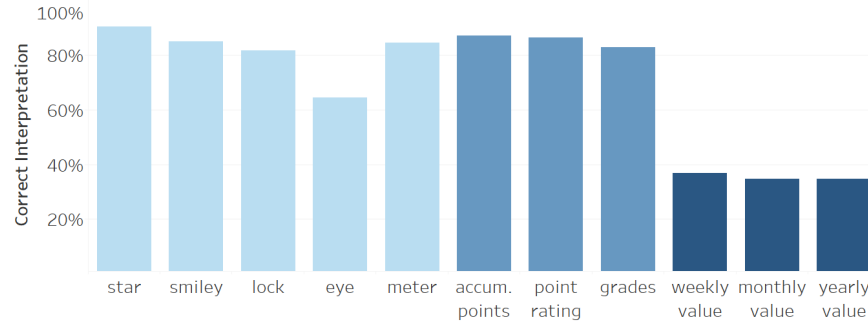


Figure 2: Percentage of participants who interpreted the privacy indicator correctly.

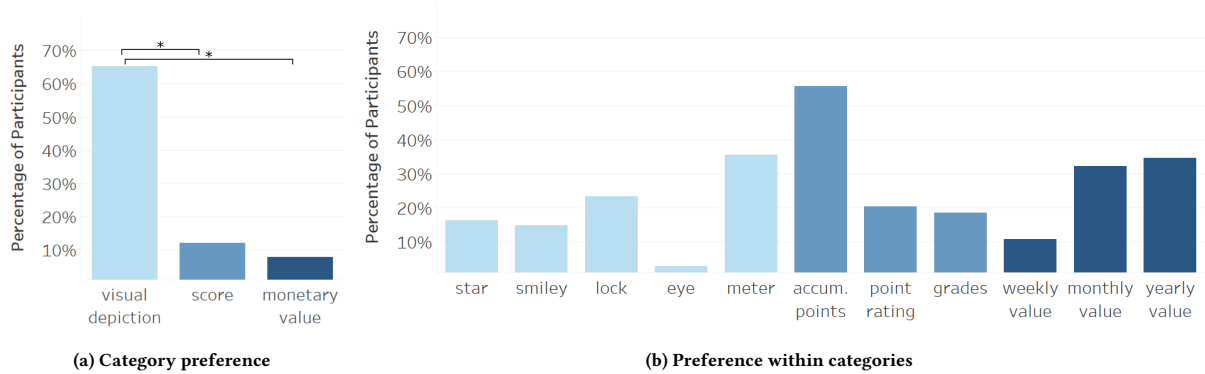


Figure 3: Preference results from Step II. The asterisk * indicates statistically significant differences with $p < 0.001$

4.1 Survey Procedure

First, we provided an *introduction and privacy policy* to the participants. Once they accepted the policy, the participants received a questionnaire for the *evaluation of the privacy indicators*. For this, we presented the participants with the different elaborations (developed in Step I, see also Figure 1) in three blocks that represent the three categories. Regarding each indicator, the participants were asked (1) what they thought this indicator represents (multiple choice) and (2) to explain their answer (open answer). At the end of each block, the participants chose their preferred privacy indicator from the different visualization possibilities. After all blocks, the participants were asked which indicator category they overall preferred. In the last step, we asked the participants to answer six questions to collect *demographics*. The full questionnaire can be found in the supplementary material.

4.2 Participants and Limitations

The survey was conducted online using LimeSurvey, which we hosted on a server in our institution. The participants were recruited via different social and professional platforms (e.g., university mailing lists, online social networks). A total of 169 participants took part in the survey, out of which 119 were full responses. 34.53% ($N = 41$) of them identified as female, 61.33% ($N = 73$) as male, 0.80% ($N = 1$) as other, and 3.34% ($N = 4$) preferred not to say. The

participants were on average 29.39 years old ($SD = 6.70$, $min = 16$, $max = 70$).

A limitation could be given because we used a convenience sample which might not be representative for all mobile users. Yet, our sample reflects young users, which, if anything, tends to reflect the demographic distribution of smartphone users [26]. We used multiple-choice items to assess the comprehensibility of indicators. This allowed us to easily compare the results of the indicators and choose those indicators that are understood correctly and preferred by the majority of participants for the subsequent lab study. Still, the answer options given have most likely primed participants towards privacy. An open answer format might have yielded different results in terms of comprehensibility, with fewer participants creating a mental link between the indicator and privacy. To address this issue, we investigated users' interpretation of the indicators using semi-structured interviews in the next step.

4.3 Study Results

In this section, we present the results from Step II based on the indicator understanding and preference.

4.3.1 Understanding of Privacy Indicators. Our results indicate that most participants correctly understood the visual depiction and the score privacy indicators (see Figure 2). However, about a third of the participants failed to interpret the visual depiction privacy indicator eye correctly. These participants mainly associated the number of

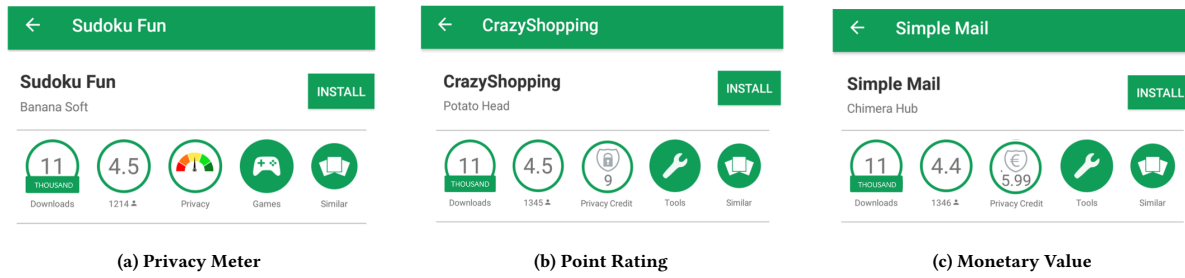


Figure 4: The three conditions used in the study.

eyes with the number of users or entities who could access their data and thus relied on a “fewer eyes equal more privacy” heuristic.

Less than 40% of our participants correctly interpreted the three monetary value indicators. Most of the remaining participants thought that the monetary value indicators specified the costs for using the app (24.0%) or the pro version of the app (22.5%). One participant associated the monetary value indicators with the weekly, monthly, or annual costs for securing their data within the app. Another participant suggested replacing the euro sign with a lock since they felt that a lock would provide a clearer association to privacy.

4.3.2 Privacy Indicator Preference. At the end of the survey, we asked the participants for their favorite indicators. Using Pearson’s chi-squared tests, we found that overall, significantly more participants preferred visual depiction privacy indicators (65.2%) compared to score privacy indicators (12.1%, $\chi^2=50.70$, $p<.001^1$), and monetary value privacy indicators (7.8%, $\chi^2=62.75$, $p<.001$, see Figure 3a).

Within the visual depiction privacy indicators, the meter was most popular (see Figure 3b). Significantly more participants preferred the privacy meter (35.5%) compared to star (16.3%, $\chi^2=9.99$, $p=.02$), smiley (14.9%, $\chi^2=11.85$, $p=.01$), and eye visualizations (2.8%, $\chi^2=39.19$, $p<.001$).

The accumulated points indicator was preferred within the score category. Significantly more participants preferred this indicator (55.6%) compared to the point rating indicator (20.4%, $\chi^2=30.56$, $p<.001$) and the grades indicator (18.5%, $\chi^2=22.87$, $p<.001$). Regarding monetary value, significantly more participants preferred to see an indication of the worth of their data per year (34.5%) or month (32.1%), compared to per week (10.7%, $\chi^2=10.53$, $p<.001$ and $\chi^2=9.00$, $p=.004$). We chose the visual meter representation, the accumulated point score, and a monetary value of 5.99 for the lab study in Step III based on these results.

5 STEP III: LAB STUDY

In the lab study, we conducted app selection tasks and semi-structured interviews with 20 participants. The participants were asked to perform four app selection tasks in four different scenarios. After completing the tasks, the participants were interviewed. The interview study served the purpose to answer the following four research questions:

RQ1: Did the participants notice the privacy indicators?

RQ2: How did participants interpret the privacy indicators?

RQ3: Which aspects did participants consider when choosing an app?

RQ4: Was the selection of the app dependent on the presence of a specific privacy indicator?

In three tasks, the participants interacted with modified Play Stores. Each modified Play Store had a badge with a privacy indicator corresponding to those chosen in Step II. By clicking on the batch, the participant received detailed privacy-related information. The fourth task served as the baseline without any privacy indicator. The modified Play Store was furthermore adapted such that we had full control of the available apps and their descriptions. The descriptions of all apps were similar, and only different based on privacy-related information.

5.1 Study Procedure

We welcomed the participants to our lab and let them read and sign a consent form. We explained to participants that they were to select apps from an app store and install them on their smartphones. The participants were asked to complete the following tasks: Select a Sudoku app, select a shopping list app, select an email app, and select an audio recording app. To avoid sequence effects, the order of these tasks was randomized using a Latin square. Participants were not informed that different privacy indicators were included in the app store for each task (for example, see Figure 4) and that one task acted as a baseline without an indicator.

After each task, the participants received a questionnaire with two items, asking them to explain which criteria they considered when choosing the app and why they chose the app they just installed. After the participants completed the tasks, we proceeded with a semi-structured interview. During the interview, we asked the participants for their favorite indicator, how they interpreted the indicator, and which aspects they consider when choosing an app. Finally, we asked the participants to provide demographics. The questionnaire and the interview guidelines can be found in the supplementary material.

5.2 Participants and Limitations

We recruited 20 participants using mailing lists, social networks, and snowball sampling. 60% of the participants identified as male, 35% as female, and 5% preferred not to say. The participants were

¹We applied Bonferroni corrections to account for type I error inflation.

on average 31.75 years old ($SD = 8.41$, $min = 21$, $max = 52$). All of them were active smartphone users.

Similarly to the online survey, the sample was rather young and thus might not be representative. Although we could determine whether the participants noticed the indicators, our results do not offer when during the app selection process, the user noticed the indicator.

5.3 Study Results

5.3.1 RQ1: Did the participants notice the privacy indicators? While participants were completing their task of selecting an app, most noticed the privacy indicators without being told by the examiner. The privacy indicator monetary value was detected by 95%, the accumulated points score by 85%, and the meter by 80% of the participants.

5.3.2 RQ2: How did participants interpret the privacy indicators? The privacy indicators were interpreted correctly with varying frequency. Most often, the meter and accumulated points were interpreted correctly. The meter was understood correctly by 70% of the participants, 25% understood it partially, and 5% did not understand it at all. The accumulated points were interpreted correctly by 70% of the participants, 30% somehow interpreted it correctly, understanding that it is an indicator, but a relation to privacy was not made. For example, they stated that accumulated points were an indicator of the app's security. Only 50% of the participants interpreted the monetary value indicator as intended. However, even if they understood the concept, they expressed confusion. For instance, P13 stated *"This indicator was confusing because at first, I thought it is the app's costs but I just clicked on it to find out what it is and it gives the rating on privacy and also tells about the permissions required for this app on your phone."* 20% of the participants understood the concept only partially. E.g., P14 stated: *"The access that the app has on my phone, credentials like contacts, storage"*, but did not mention the monetary aspect. 30% of the participants did not understand the privacy indicator monetary value correctly. For instance, P19 said: *"Maybe I have to pay for it if I want to make it safer."*

5.3.3 RQ3: Which aspects did participants consider when choosing an app? Overall, in 65% of app decisions, participants mentioned privacy as a decision criterion. Many participants who chose the most privacy-friendly app indicated that privacy was a deciding factor in their app selection process. However, there were also differences between the privacy indicators. While 93% of the participants named privacy as a decision criterion when selecting the shopping app (privacy indicator: accumulated points), it was 83% for the Sudoku app (privacy indicator: meter), and only 64% for the email app (privacy indicator: monetary value). When selecting the shopping app (accumulated points), several other decision criteria were mentioned, such as the number of downloads, ratings, reviews, the user interface, or the vendor. Various factors were also considered when selecting the email app (monetary value), such as rating, number of downloads, vendor, or user reviews. When selecting the Sudoku app (meter), the app's rating was usually the only other factor mentioned.

5.3.4 RQ4: Was the selection of the app dependent on the presence of a specific privacy indicator? The most privacy-friendly app was chosen most frequently by participants who saw the privacy meter (90% of the participants), followed by accumulated points with 80% and monetary value with 70%. In the baseline task (no privacy indicator), all apps were selected with roughly the same frequency.

6 DISCUSSION AND OUTLOOK

With this study, we investigated simple privacy indicators for supporting users when installing new mobile apps. For this, we first developed three categories of privacy indicators (visual depiction, score, and monetary value), each containing diverse visualization possibilities based on the scientific literature. In our second step, we evaluated the privacy indicators in terms of comprehensibility and preference through an online survey. Building on these results, in the final step, we investigated how users perceive the various privacy indicators in an app store and to which extent the indicators influence the users' decision to install an app. For this, we conducted a lab study with 20 participants.

Our work differs from existing research in the following aspects. Previous work mostly focused on placing additional privacy-related information in app stores, such as examples of data being shared [13] or statistical information [17]. Alternatively, privacy-related information can be aggregated to support users' comprehensibility, e.g., in the form of scores. We are building on the scores approach by exploring simple ways to communicate these scores to users.

6.1 Study Results

Our study results indicate preference of visual depiction in the form of a privacy meter. Even though this indicator was not always discovered directly, it was most often interpreted correctly in the lab study and most often led to participants choosing the most privacy-friendly app. A possible explanation for why most participants preferred the meter could be that the privacy-related information is highly aggregated and quickly comprehensible. The other indicators might require previous knowledge to understand them completely. E.g., for understanding the accumulated points, knowledge of minimal and maximal values are required. Further, the (colorful) design is visually most appealing for the participants. Monetary value was often not understood correctly. This could be explained by the lack of familiarity of many users with the concept of "value of data" and the fact that collecting the information requires more (reading) effort.

6.2 Recommendations

Based on our investigations, we provide the following recommendations for simple privacy indicators.

1) Comprehensibility: Privacy indicators should be designed in a way to be easily noticed and understood by users. Ideally, the users would not require any previous knowledge for understanding the indicator. We found that the concept of assigning a monetary value to the data shared via an app was confusing for most participants. Thus, this concept seems not yet ready to be implemented as a privacy indicator, and more elaborate solutions are needed to

communicate to users how and when they "pay" for an app with their data. Although scores can be used as a basis for information, they should be presented to users in a way that they can be understood quickly (e.g., in the form of a visual depiction).

2) Unique Interpretation: The interpretation of the indicators should be unique, and the representation should be distinguishable from other icons in the app store. For this purpose, it may be useful to use representations that users are familiar with from other contexts for evaluating products. The greatest challenge here will be to find an implementation that is both intuitively interpretable and related to privacy. The eye indicator, e.g., relates to users' mental models of privacy, indicating that they are watched. Still, our participants struggled with interpreting this indicator, as more eyes are intuitively associated with less privacy. In contrast, more items in a visual scale, e.g., in a star rating, usually indicate the best expression of the represented feature.

3) Information Gateway: Privacy indicators can only present aggregated information. Hence, they should serve as a gateway for users to get more information on-demand. In our lab study, users clicked on the indicator and read the information behind it.

7 CONCLUSION & FUTURE WORK

Privacy indicators should be as simple, clearly understandable, and as easily recognizable as possible, so that they help users select privacy-friendly apps that match their privacy preferences. Visual depictions, such as a privacy meter, seem to be particularly suitable for this purpose. Based on our investigations, we consider the following aspects as integral steps for future work. In our studies, we investigated the monetary value of the collected data as a privacy indicator because related work had demonstrated that people are willing to pay for premium services based on privacy indicators [11, 27]. Both studies demonstrated issues regarding the comprehensibility of our depictions. Future studies should investigate methods to provide easy-to-understand indicators based on monetary value. A promising solution for that might be presenting the comparison of the cost of a premium version and the value of the collected data. In our study, we investigated whether participants noticed the indicators by asking them. An eye-tracking study might offer valuable information about the point of time a user notices an indicator or how long they look at it.

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